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Experiment No.: 1

Aim: Study and Implementation of Basic Git and GitHub Operations.

This Experiment maps to LO2.

LO2: To understand and perform the basic version control operations using **Git** and **GitHub**, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory:

Git is a **Distributed Version Control System (DVCS)** developed by Linus Torvalds. It helps developers track changes made to files, maintain multiple versions of a project, collaborate with team members, and restore previous versions when necessary.

GitHub is a cloud-based platform that hosts Git repositories, allowing developers to collaborate, manage projects, review code, and maintain version history.

Using Git and GitHub enables efficient project management, supports collaborative software development, and ensures data integrity through version control.

Procedure / Steps Performed

1: Configure Git Username

Configured the Git username.

git config --global user.name "Janhvi-12"

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2. Configure Git Email

Configured the Git email address.

git config --global user.email "ramrakhyanijanhvi@gmail.com"

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3. Cloned the Repository

Cloned the remote repository to the local system using:

git clone https://github.com/Janhvi-12/Git-Practice.git

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4. Checked Repository Status

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Verified the repository status using:

git status

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5. Created a New File

Created a file named **hello.txt** and added sample text.

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6. Committed Changes

Saved the staged changes to the local repository using:

git commit -m "Added hello.txt"

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7. Pushed Changes

Uploaded the local commit to GitHub using:

git push

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8. Pull Changes
Downloaded the latest changes from GitHub.

git pull

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10. View Branches

Displayed the list of local branches.
git branch

---

10. Created a New Branch

Created a new branch named **feature** using:

git branch feature

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11. Switched to the Branch

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Changed the working branch using:

git checkout feature

---

12. Merged Branch

Returned to the **main** branch and merged the **feature** branch using:

git merge feature

The merged changes were then pushed to GitHub.

13: View Remote Repository

Displayed the configured remote repository.

git remote -v

14: View Git Configuration

Displayed the Git configuration.

Command:

git config –list

15: View Commit History

Displayed the complete commit history.

git log

16: View Commit History (One Line)

Displayed the commit history in one-line format.

Command:

git log --online

17: View All Branches

Displayed all local and remote branches.

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Command:

git branch -a

18: Fetch Changes

Fetched changes from the remote repository.

Command:

git fetch

19: View Remote Details

Displayed detailed information about the remote repository.

Command:

git remote show origin

20: View Latest Commit

Displayed details of the latest commit.

Command:

git show

21: Switch Branch (git switch)

Switched to another branch using git switch.

git switch testing

22: Rename Branch

Renamed the current branch.

git branch -m testing demo

23: Delete Branch

Deleted the local branch.

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git branch -d demo

24: Rename File

Renamed a tracked file.

git mv hello.txt greetings.txt

25: Remove File

Removed a tracked file from the repository.

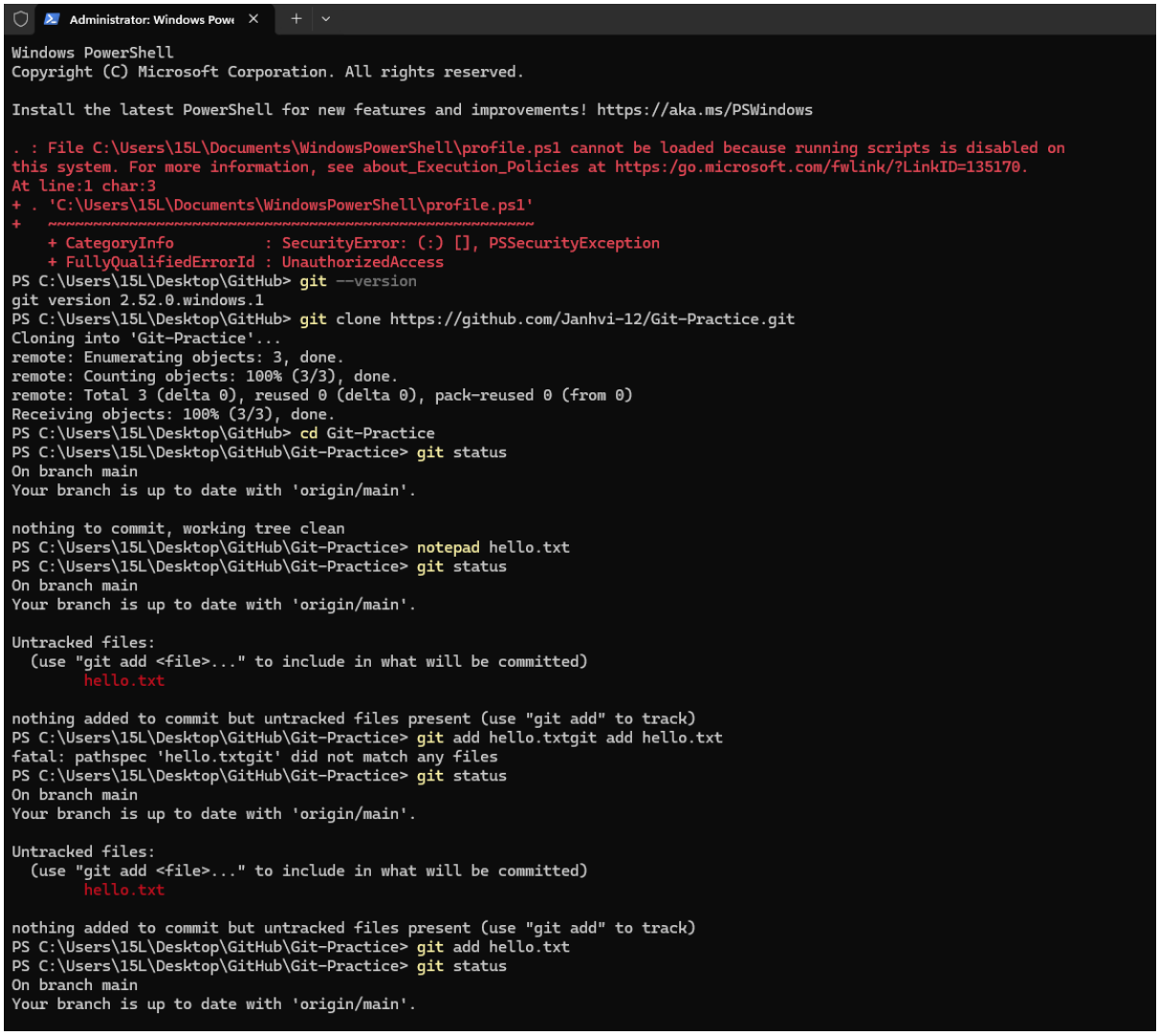
git rm greetings.txt

26: Authorization / Git Configuration Verification

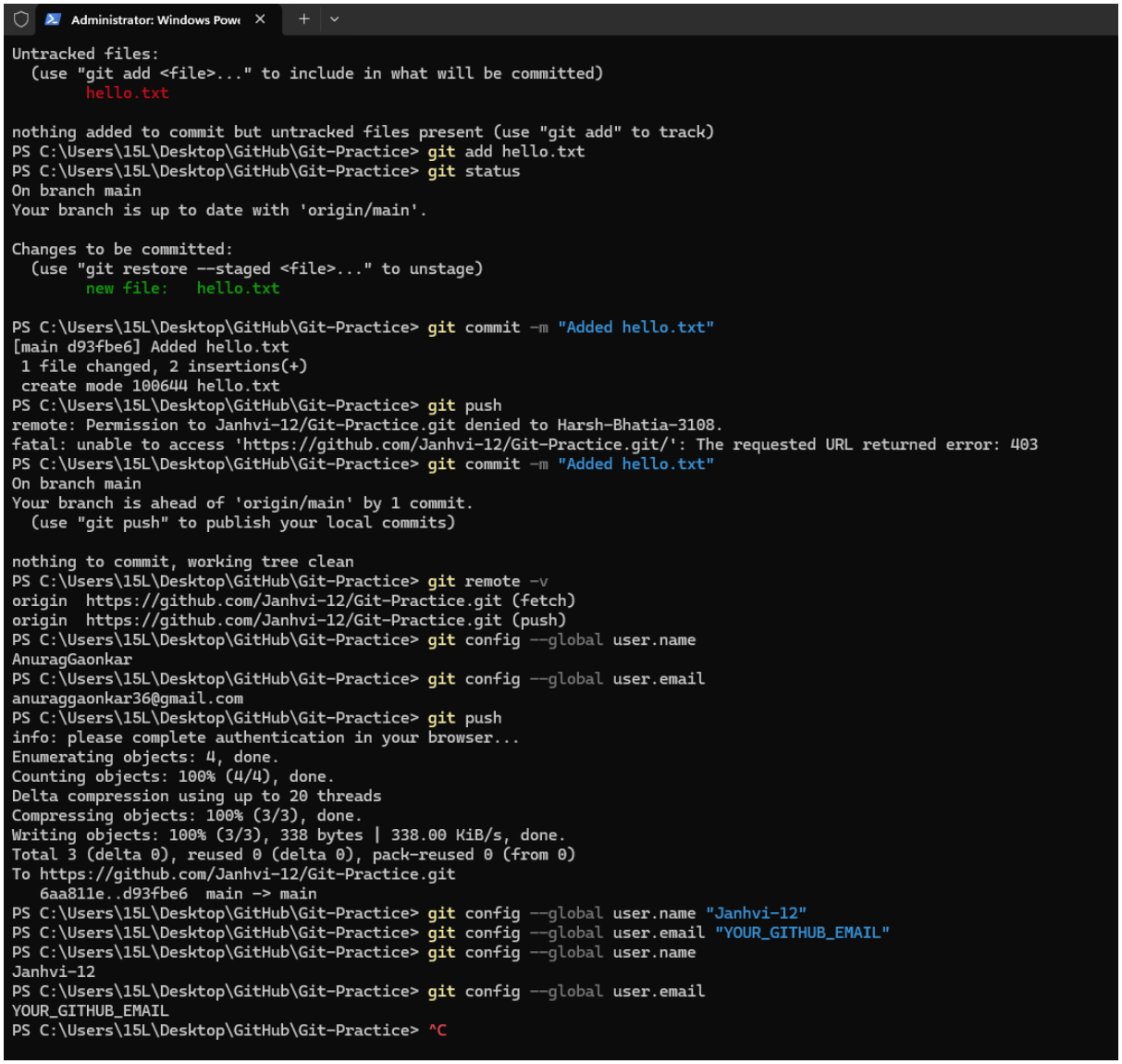
Verified the Git configuration and authorization settings.

git config --list

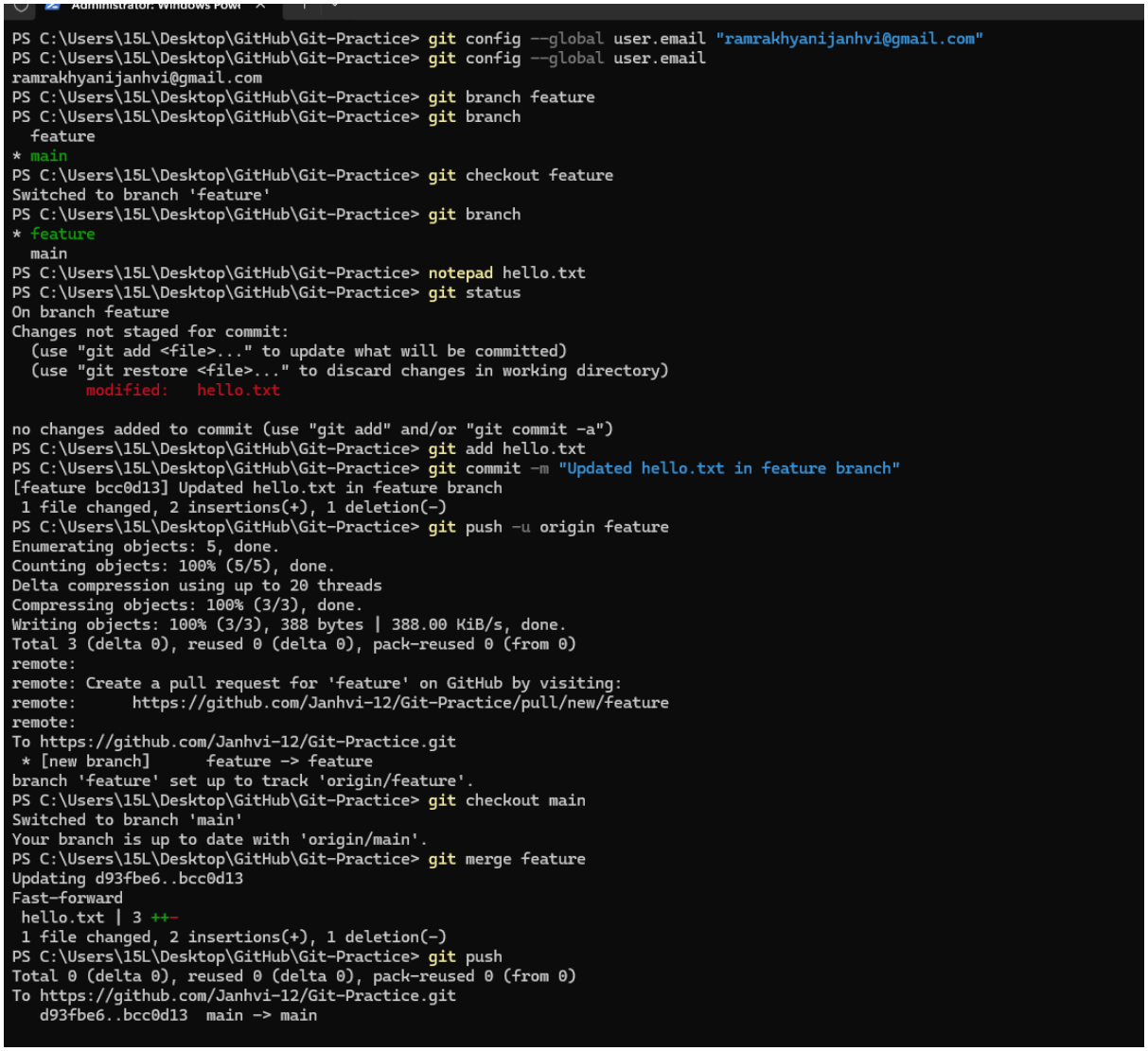
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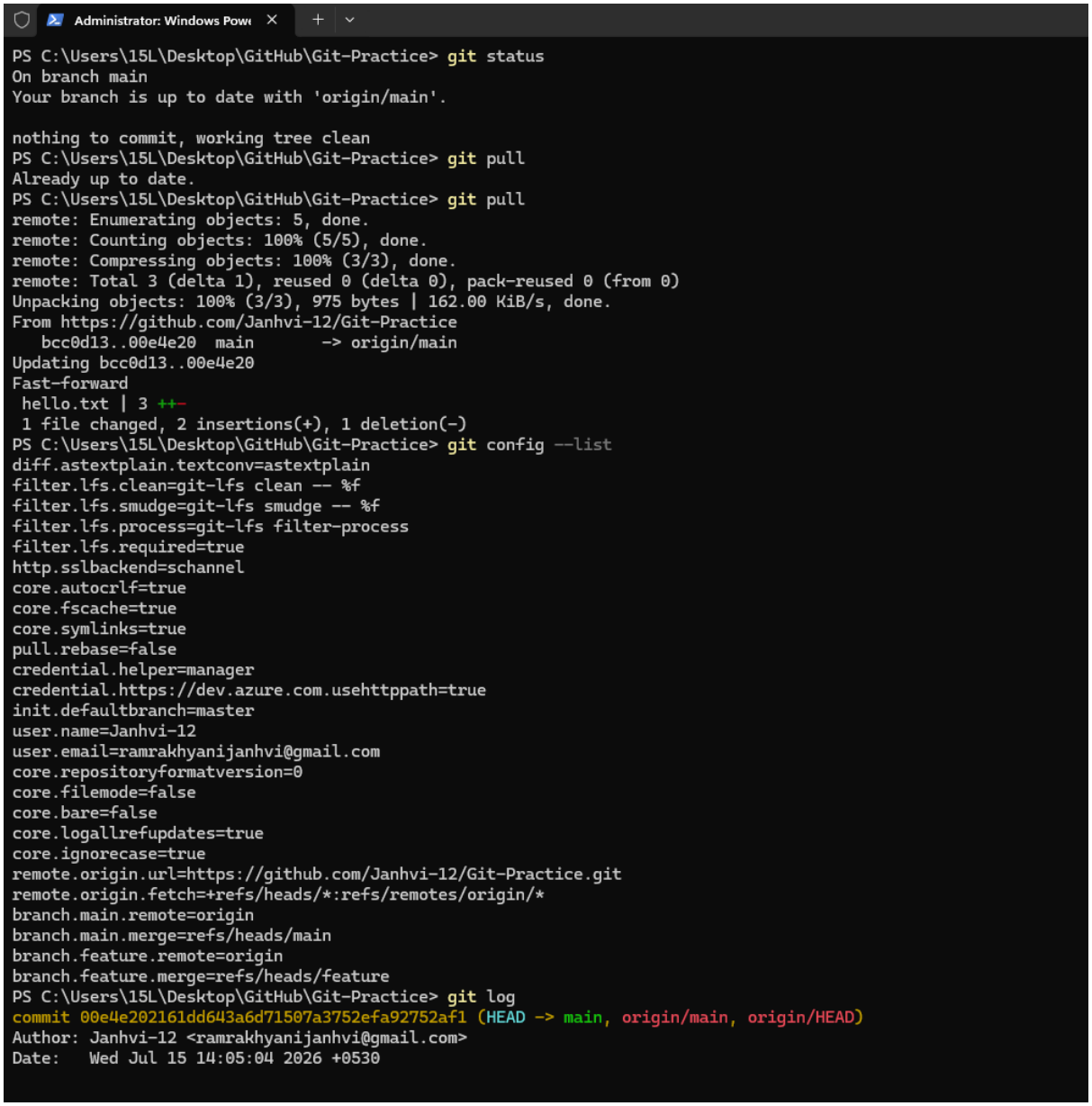
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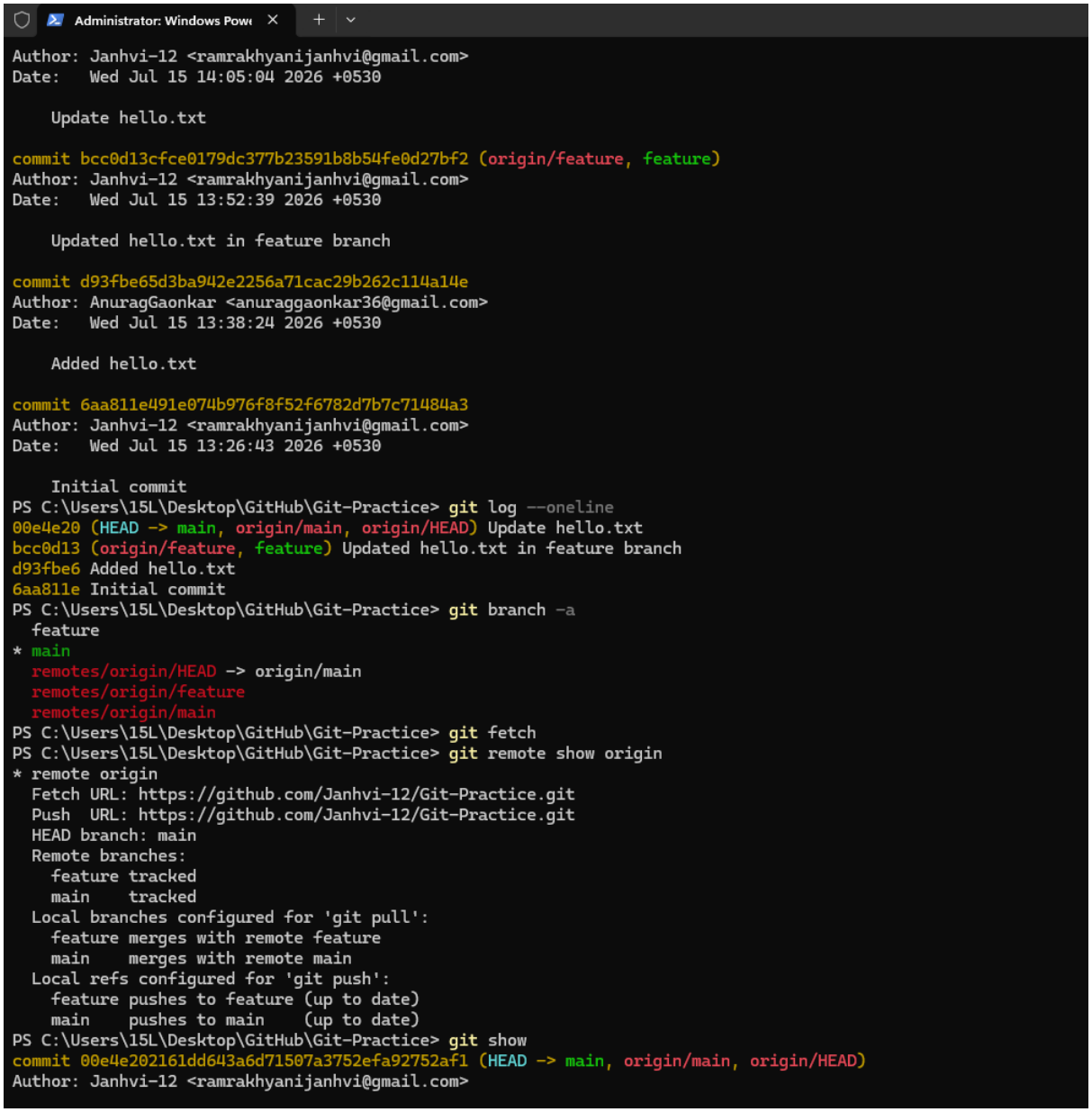
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```
commit 00e4e202161dd643a6d71507a3752efa92752af1 (HEAD -> main, origin/main, origin/HEAD)
Author: Janhvi-12 <ramrakhyanijanhvi@gmail.com>
Date:   Wed Jul 15 14:05:04 2026 +0530

    Update hello.txt

diff --git a/hello.txt b/hello.txt
index c992aa2..bcc348b 100644
--- a/hello.txt
+++ b/hello.txt
@@ -1,3 +1,4 @@
 Hello, this is my first Git file.
 I am learning Git and GitHub.
-This line was added in the feature branch.
\ No newline at end of file
+This line was added in the feature branch.
+Added directly from GitHub.
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git branch testing
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git switch testing
Switched to branch 'testing'
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git branch -m testing demo
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git branch -d demo
Deleted branch demo (was 00e4e20).
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git mv hello.txt greetings.txt
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git rm greetings.txt
error: the following file has changes staged in the index:
    greetings.txt
(use --cached to keep the file, or -f to force removal)
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git commit -m "Deleted greetings.txt"
[main e06e680] Deleted greetings.txt
 1 file changed, 0 insertions(+), 0 deletions(-)
 rename hello.txt => greetings.txt (100%)
PS C:\Users\15L\Desktop\GitHub\Git-Practice>
```

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```
PS C:\Users\15L\Desktop\GitHub\Git-Practice> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=true
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Janhvi-12
user.email=ramrakhyanijanhvi@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
remote.origin.url=https://github.com/Janhvi-12/Git-Practice.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
branch.feature.remote=origin
branch.feature.merge=refs/heads/feature
PS C:\Users\15L\Desktop\GitHub\Git-Practice>
```

Commands Used

Sr. No.	**Command**	**Purpose**
1	git config --global user.name	Configure Git username
2	git config --global user.email	Configure Git email
3	git clone	Clone remote repository
4	git status	Check repository status
5	git add	Stage files
6	git commit	Save changes locally
7	git push	Upload commits to GitHub
8	git pull	Download and merge changes
9	git branch	Create/List branches
10	git checkout	Switch branches
11	git merge	Merge branches

12	git remote -v	View remote repository
13	git config --list	Display Git configuration
14	git log	View commit history
15	git log --oneline	View concise commit history
16	git branch -a	List all branches
17	git fetch	Download changes without merging
18	git remote show origin	View remote repository details
19	git show	Display latest commit details
20	git switch	Switch branches (alternative to checkout)
21	git branch -m	Rename a branch
22	git branch -d	Delete a local branch
23	git mv	Rename a tracked file
24	git rm	Remove a tracked file
25	git tag	Create/List repository tags

Conclusion:

This experiment provided practical experience in using Git and GitHub for version control and repository management. Various Git operations, including repository configuration, cloning, staging, committing, pushing, pulling, branching, switching branches, and merging, were successfully performed.

The experiment also demonstrated the synchronization between local and remote repositories and highlighted the importance of version control in software development. Overall, the practical enhanced understanding of Git workflows, repository management, and collaborative development